Wildlife Ecology

Week 2 Assignment Solutions (2026)

1. Hierarchy emerges almost inevitably through a wide variety of evolutionary processes, for the simple reason that hierarchical structures are _____ (Fill in the blank)

a. perfect

b. imperfect

c. **stable**

d. unstable

2. Which level is formed when similar cells and their extracellular matrix from the same origin together perform a specific function?

(a) Organ

(b) **Tissue**

(c) Organ system

(d) Population

3. "the diversity that exists within an ecosystem" is

a. **alpha (α) biodiversity**

b. beta (β) biodiversity

c. gamma (γ) biodiversity

d. delta (δ) biodiversity

4. "groups of actually or potentially interbreeding natural populations, which are reproductively isolated from other such groups" is a definition of

a. cells

b. **species**

c. ecosystems

d. biomes

5. The emergent principle can be stated as

a. Whole = sum of parts

b. Whole < sum of parts

c. **Whole > sum of parts**

d. None of these

6. The smallest unit of life is the:

(a) **Cell**

(b) Tissue

(c) Organ

(d) Organism

7. The diversity of genetic information present at the levels of phyla, families, species, populations and individuals is called:

(a) Ecosystem biodiversity

(b) Species evenness

(c) **Genetic biodiversity**

(d) Species richness

8. The value of leaving use and non-use values for future generations is called:

(a) Altruistic value

(b) Option value

(c) Existence value

(d) **Bequest value**

9. Value arising even though a resource is not being used is called:

(a) Direct value

(b) Indirect value

(c) **Non-use value**

(d) Productive value

10. Species richness refers to:

(a) The distribution of individuals among different species

(b) The genetic variation within one population

(c) The number of ecosystems in a biome

(d) **The number of species present**